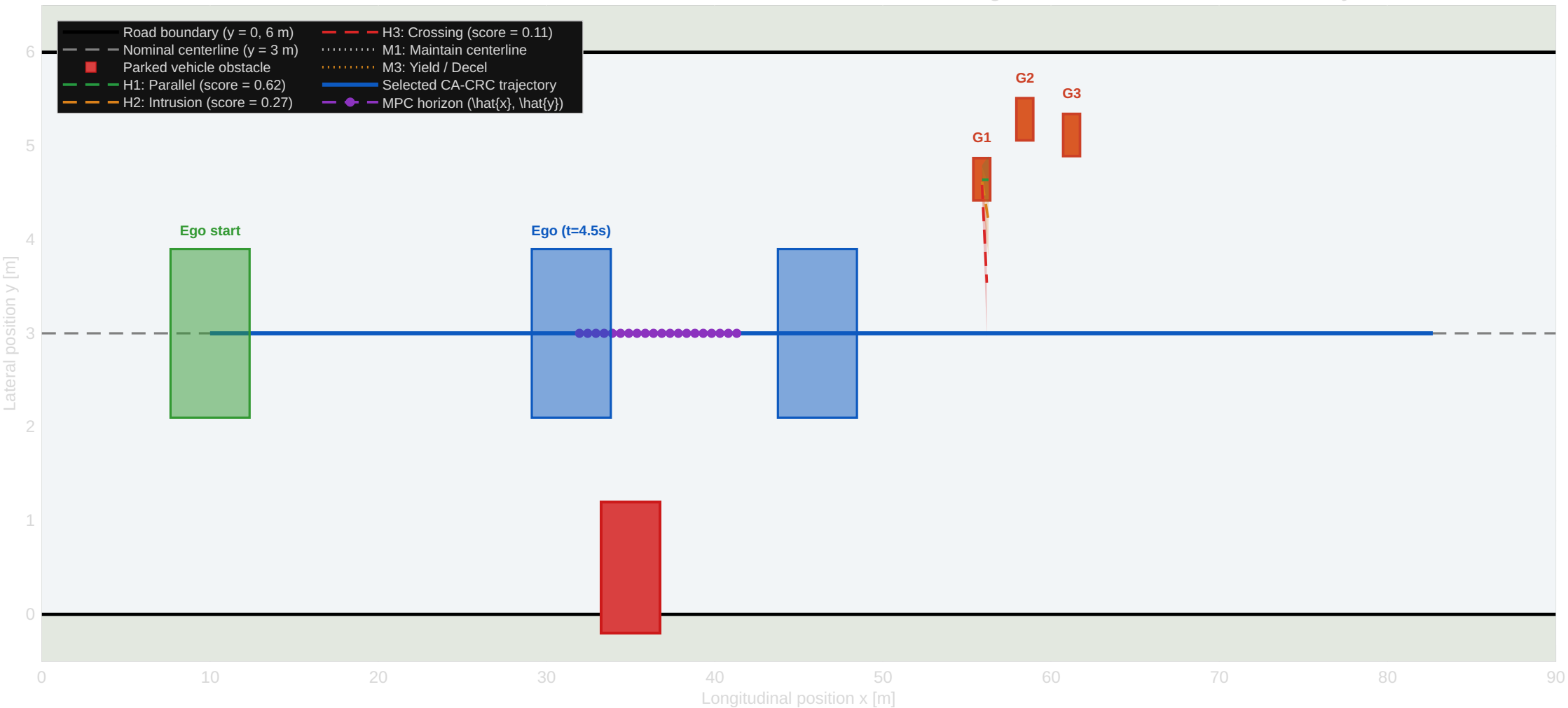




(a) Perception & Tracking State

Time step $t = 4.4 \text{ s}$
 Tracked Agents: 3 Livestock
 Ego State: $(31.5\text{m}, 3.0\text{m}, 4.9\text{m/s})$

Agent	x [m]	y [m]	vx [m/s]	vy [m/s]	Status
G1	55.9	4.64	0.20	-0.15	INTRUDING
G2	58.4	5.28	0.10	-0.05	PARALLEL
G3	61.2	5.11	0.05	-0.02	GRAZING

(b) Multi-Hypothesis Prediction

Intrusion Agent G1 ($x=55.9\text{m}, y=4.64\text{m}$)
 Lookahead Horizon $N_p = 2.0 \text{ s}$

Candidate Hypotheses:
 H1 (Parallel Continuation): Score = 0.62
 H2 (Road Encroachment): Score = 0.27
 H3 (Sudden Crossing): Score = 0.11

* Scores derived from kinematic heading and road alignment model.

(c) Planning & Safety Filter

Ego Candidate Maneuvers:
 M1 Maintain Lane: Cost = 8.50 (High Risk)
 M2 Cautious Shift: Cost = 1.25 (OPTIMAL)
 M3 Emergency Yield: Cost = 4.80 (Conservative)

Selected Maneuver: M2 (CA-CRC MPC)
 Safety Filter Status: PASS (Constraints satisfied)